Ruoxian Zhang

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EDUCATION

University of Illinois Urbana-Champaign (GPA: 4.0), Illinois, USA Expected 05/2027
M.S. in Kinesiology
Graduate Research Assistant (Fully Funded)

Beijing Sport University (Major GPA: 4.0; GPA: 3.91), Beijing, China 09/2021-07/2025
B.S. in Strength and Conditioning
B.Ed. in Physical Education

Springfield College (GPA: 4.0), Massachusetts, USA 09/2024-12/2024
Exchange Student (full-ride scholarship)

PUBLICATIONS & PRESENTATIONS

Yue, C., Xiong, W., Zhang, C., **Zhang, R**., Deng, K., & Li, Z. (2025). The effects of accentuated loading repetition structures on muscle strength and adaptations. *The Journal of Strength and Conditioning Research, 39*(12), 1244-1253.

Zhang, R., Witte, P., Malani, R. N., Ji, F., Keye, S. (Submitted). Evaluation of AI models for Movement Assessment: Agreement with Experts Using Raw and Pose- Based Inputs. *The Journal of Healthcare Informatics Research.*

Zhang, R., & Ji, F. (In preparation). A tutorial on agreement between AI and human ratings of fundamental movement skills. Manuscript in preparation for submission to *Measurement in Physical Education and Exercise Science.*

Kondagunta, P., Witte, P. K., **Zhang, R.**, Malani, R. N., & Keye, S. A. (2026, April). *MIND: Mind-body intervention and neural dynamics—A randomized crossover protocol* [Conference presentation]. University of Illinois Urbana-Champaign Undergraduate Research Symposium, Urbana, IL, United States.

Edwards, M., Witte, P. K., **Zhang, R.**, Malani, R. N., & Keye, S. A. (2026, April). *Weight, menopause, and movement: Predictors of femoral neck bone density in U.S. women* [Conference presentation]. University of Illinois Urbana-Champaign Undergraduate Research Symposium, Urbana, IL, United States.

Zhang, R., Witte, P. K., Malani, R. N., Ji, F., & Keye, S. A. (2026, April). *Evaluation of generative AI models for movement assessment: Agreement with experts across LLMs using raw and pose-based inputs* [Conference presentation]. 2026 Midwest ACSM Mid-Year Virtual Symposium. Virtual.

RESEARCH EXPERIENCE

Research Enhancing Academic Development in Youth Project (READY Project)
Research Assistant, PLAY Lab, University of Illinois Urbana-Champaign 08/2025-present
PI: Dr. Shelby Keye

- Supported participant testing and EEG data collection for a longitudinal study of motor, cognitive, and academic development.
- Preprocessed EEG data using EEGLAB/ERPLAB in MATLAB, including data reduction and bad-channel removal.
- Generated grand-average ERP waveforms and topographic plots for group-level interpretation and lab presentations.
- Trained undergraduate research assistants in study procedures and participant testing protocols.

Fitness and Activity for Motor and Executive Function (FAME) Study
Research Assistant, PLAY Lab, University of Illinois Urbana-Champaign 08/2025-present
PI: Dr. Shelby Keye

- Conducted participant testing and supported fNIRS setup and data collection for a multimodal study of motor, cognitive, and physical function.
- Independently processed markerless motion-capture data using Theia3D for all participants across 13 movement tasks.
- Generated task-related fNIRS brain maps and developed visualization instructions shared with lab colleagues.
- Reviewed literature on motor development, executive function, and physical activity interventions.

Generative AI for movement assessment in physical education and kinesiology (MOVE Project)
Research Assistant, University of Illinois & University of Toronto 08/2024-present
Supervisors: Dr. Shelby Keye & Dr. Feng Ji

- Led an interdisciplinary project evaluating multimodal large language models for image-based movement assessment.
- Evaluated agreement between AI-generated and expert movement scores using raw-image and pose-based inputs.
- Designed and refined prompting and input strategies to optimize model performance.
- Led manuscript preparation from research question development through journal submission.

Adolescent physical activity and mental health using the Future of Families & Child Wellbeing Study
Research Assistant, AI and Prevention Science Lab, University of Florida 07/2024-05/2025
PI: Dr. Xiaoya Zhang

- Reviewed literature and summarized findings on family attitude, motivation, physical activity, and mental health among adolescents.
- Assisted in data cleaning and management.
- Conducted data analysis including descriptive statistics, t-test, ANOVA, correlation analysis, and regression.

Exploring the utility of wearable device in promoting physical activity motivation
Undergraduate Thesis, Beijing Sport University & Springfield College
Mentors: Drs. Susan Guyer & Zezhao Chen 06/2024-01/2025

- Reviewed literature on exercise motives, physical activities, and wearable device among college students.
- Designed online quantitative survey.
- Collected data among college students in Beijing, China.
- Analyzed data to explore the role of wearable device in increasing physical activities and exercise motives using independent sample t-test.

The effects of accentuated loading repetition structures on muscle strength and adaptations
Research Assistant, Biomechanics Lab, Beijing Sport University 2023-2024
Mentor: Caizheng Yue

- Designed and conducted an eight-week training intervention to promote sport performance.
- Facilitated pre- and post-test data collection using ultrasound-based measurements in a lab setting with college athletes.
- Conducted descriptive data analysis, paired t-test to evaluate the effects of the intervention on multiple sport performance markers.
- Contributed to manuscript preparation resulting in a peer-reviewed publication in the Journal of Strength and Conditioning Research.

Exploring the influence of musical rhythm on preschool children's executive function
Research Coordinator, Springfield College *08*/2024-12/2024
Mentor: Bingbing Li

- Facilitated tasks to measure children's executive function.

TEACHING EXPERIENCE

Teaching Assistant, University of Illinois Urbana-Champaign 08/2026-05/2027

HK 340: Sociology & Psychology of Physical Activity (Laboratory Instructor)

- Lead 2 weekly discussion and laboratory sections (~50 students total), facilitating student engagement with social-psychological concepts in physical activity.
- Guide interactive class discussions, supervise lab activities, and evaluate student academic performance.

HK 102: Bowling (Primary Instructor)

- Serve as the primary instructor for a full class section (40 students).
- Design course progression, teach motor skill execution.

HK 271: Health Equity in the United States (Grader)

- Evaluate and grade undergraduate coursework, assignments, and exams.
- Provide detailed constructive feedback on health disparity topics and student writing.

HONORS & AWARDS

OEAI (Certified eligibility for Teaching Assistantship), UIUC	2025
Certified Strength and Conditioning Specialist (CSCS)	2025
Hsu Scholarship for Exchange Program at Springfield College ($17,000)	2024
Beijing Sport University Scholarship for Academic Excellence (¥1,500)	2023
First Prize in Competition of Basic Skills for Physical Education Teaching	2023
First Place of Shandong Provincial Sports Exam (1st among 25,000 examinees)	2021

WORK EXPERIENCE

Volunteer 11/2024
Mile 13 Shift 1, 2024 TCS Marathon, New York City, USA
Intern 07/2024-09/2024
National Training Base, Beijing Sport University
Strength and Conditioning Coach 02/2024-06/2024
Ice Hockey U18 Team, Beijing Sport University
Track & Field Coach 08/2023-10/2023
100-meter, Guangdong Provincial Athletics Team

SELECTED COURSEWORK

Research Methods/Statistics: Research methods in health and kinesiology, Quantitative data management, Educational statistics, Scientific research methods, Sport statistics
Substantive Topics: Brain health, Biomechanics, Sport psychology, Sport physiology, Physical education pedagogy, Basic principles of human movement, Strength and conditioning, Motor learning and control, Sport anatomy, Sport injury, Movement and sports studies
Coursera/DataCamp: R programming, GPT program engineering, R for data science, Data visualization

SKILLS

Quantitative Methods: Generalized linear models (linear, logistic, Poisson), Multivariate regression and model diagnostics, Longitudinal and repeated-measures analysis, Hypothesis testing and statistical inference, Data cleaning, preprocessing, visualization, and reproducible analysis workflows
Programming & Statistical Software: R (data wrangling, modeling, visualization), SPSS, Minitab, Google Colab
Biomechanics & Motion Analysis: Theia3D, Visual3D
Neurophysiological & Experimental Tools: EEGLAB/ERPLAB (MATLAB), Brainstorm, Homer3, E-Prime, NIRS Toolbox
Other Technical Tools: LaTeX, Microsoft Office Suite